



UVAN SHANKAR S J

Software Engineer with 2 years of experience in test automation, specializing in Python and Ruby. Skilled in developing automation frameworks, enhancing testing efficiency, and collaborating with cross-functional teams to deliver high-quality software.

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SKILLS & LANGUAGES:

Languages: Python (Advanced - OOP, Threading, Unit Testing) , JavaScript (ES6+), Ruby, SQL, C++, HTML, CSS.

Databases: MySQL (Sequelize ORM), MongoDB, DynamoDB.

Front-End: React.js (Context API, State Management, Lazy Loading,redux store), Tailwind CSS.

Back-End: Node.js, Express.js, REST API Design, JWT Authentication.

Cloud: AWS (EC2, S3, Lambda), GitHub, SonarQube, Sentry, Jest,Jenkins (CI/CD), Perforce (Helix Core),Docker (Exposure)

Automation: Selenium, Integration Testing, Computer Vision-based Test Tools

Operating Systems: Linux (Ubuntu)

Soft Skills: Leadership, Communication, Teamwork, Problem Solving, Critical Thinking.

WORK EXPERIENCE:

Software Development Engineer in Test 1

ICU Medical ,Chennai |Jan 2024 - Present:

- Led the development of computer vision-based automation tools for testing infusion pumps, reducing manual effort and improving feedback cycles.
- Created 20+ Python functions to automate workflows and manage internal data across devices.
- Developed **Bash shell scripts** to automate environment configuration and dependency management on **Linux-based systems**, reducing developer onboarding time by 30% .
- Built **Mednet**, a centralized web dashboard for monitoring hospital devices.
- Automated log file management, reducing errors and saving significant manual time
- Developed python script uses the API to migrate,edit and resume the test case much more efficiently which reduces the manual effort and error.
- Managed complex codebases and hardware-software integration scripts using **Perforce**, ensuring version integrity across multi-functional engineering teams
- Collaborated in an **Agile environment** using a **concurrent development and testing flow**, ensuring critical functionality was prioritized and validated in parallel with ongoing development to reduce time-to-market.

PROJECTS:

Expansion - (2025)([Link](#))

- Developed the FE and BE for a expenses tracking website with functionalities like
- Implemented React concepts like Lazy Loading, Context API, State Management.
- Implemented mongo DB, REST APIs for the backend.

Faculty management system— (2023) ([link](#))

- Built a MERN stack (MySQL, Express.js, React.js, Node.js) dashboard for workload tracking.
- Increased faculty productivity by 40% through streamlined process automation.
- Used Sequelize ORM and session management with JWT for secure data handling.
- Integrated API testing, performance testing, and front-end unit tests with tools like Selenium, Jest, and SonarQube.

Autonomous delivery robot— (2023) ([link](#))

- Created a web interface for sending/receiving robot navigation data via REST API.
- Integrated AWS cloud database services DynamoDB into project architecture to ensure scalable and secure data storage.
- Developed an instruction-based path delivery system; future work includes optimizing routes using ride data.

EDUCATION:

B.Tech in Computer Science and Engineering

Amrita Vishwa Vidyapeetham, Coimbatore – 2020 to 2024

CGPA: 7.31/10

Certification:

AWS Cloud Computing (Introduction to Cloud): ([link](#))

Learned and implemented the AWS cloud services like S3, EC2, Lambda, DynamoDB, Cloud trial, Cloud watch, IAM, CloudFront, and ElastiCache.

Additional:

Solved 130+ programming challenges on LeetCode using Python ([link](#))

Strong foundation in Data Structures, Algorithms, and Backend System Design.